

# Kitchen Pi Magic Mirror

*Full Setup Guide — YouTube Recording Reference*

Picking up after Raspberry Pi OS is flashed and booted

## 1. MagicMirror<sup>2</sup> — Initial Installation

 Show the terminal. Walk through each command, explain what it does before you run it.

### 1.1 Install Node.js (via NVM — recommended)

MagicMirror requires a recent version of Node. The cleanest way on a Pi is via NVM so you can upgrade Node later without breaking things.

```
curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.39.7/install.sh | bash
source ~/.bashrc
nvm install --lts
node -v && npm -v
```

 Confirm both node and npm respond with version numbers before moving on.

### 1.2 Clone and Install MagicMirror

```
git clone https://github.com/MagicMirrorOrg/MagicMirror
cd MagicMirror
npm run install-mm
```

 This takes a few minutes on a Pi. Good time to explain what MagicMirror actually is for anyone new to the project.

### 1.3 Copy the Sample Config

```
cp config/config.js.sample config/config.js
```

This gives you a working baseline. Everything you configure from here will be edits to this file.

### 1.4 Start MagicMirror for the First Time

```
npm start
```

You should see the default MagicMirror screen — clock, calendar placeholder, compliments. If it loads, the base install is good. Quit with Ctrl+C and we'll configure it next.

### 1.5 Set MagicMirror to Start on Boot (PM2)

```
npm install -g pm2
pm2 start npm --name "magicmirror" -- start
pm2 save
pm2 startup
```

Run the command PM2 outputs — it will look like a sudo env ... command. This registers PM2 with systemd so MagicMirror survives reboots.

 Test it: `sudo reboot`, wait for it to come back up, and confirm MagicMirror auto-launched.

## 2. Installing Modules

 Show the `modules/` directory. Explain that every third-party module lives here as its own folder, and is git cloned in then `npm` installed.

### 2.1 MMM-CalendarExt3

Replaces the default calendar module with a proper weekly/monthly grid view. This is what fills the entire bottom half of the mirror.

```
cd ~/MagicMirror/modules
git clone https://github.com/MMRIZE/MMM-CalendarExt3
cd MMM-CalendarExt3 && npm install
```

 The default calendar module stays in `config.js` because `CalendarExt3` reads from it as a data source — you'll see how they're linked in the config section.

### 2.2 MMM-BackgroundSlideshow

Displays a rotating background image behind the mirror content. Images are stored locally on the Pi.

```
cd ~/MagicMirror/modules
git clone https://github.com/darickc/MMM-BackgroundSlideshow
cd MMM-BackgroundSlideshow && npm install
```

Create your images folder and drop your photos in:

```
mkdir -p ~/MagicMirror/modules/MMM-BackgroundSlideshow/kitchenpiimages
```

 Copy your images into that folder via `scp`, USB, or Google Drive — however you prefer.

### 2.3 MMM-MyScoreboard

Shows live and recent scores for NHL, NBA, NFL, MLB, NCAAM, and NCAAF for your chosen teams.

```
cd ~/MagicMirror/modules
git clone https://github.com/jclarke0000/MMM-MyScoreboard
cd MMM-MyScoreboard && npm install
```

### 2.4 MMM-YouTubeSubscribers

Displays a live YouTube subscriber count. Requires a Google API key.

```
cd ~/MagicMirror/modules
git clone https://github.com/Kresss/MMM-YouTubeSubscribers
cd MMM-YouTubeSubscribers && npm install
```

You'll need a YouTube Data API v3 key from Google Cloud Console. We'll add it to the config in the next section.

### 2.5 Built-in: Weather Module

The weather module ships with MagicMirror — no git clone needed. We're using the openmeteo provider which requires no API key, just lat/lon coordinates.

## 3. Configuring config.js

 Open config/config.js in nano or your editor of choice. Walk through each module block. Replace \*\*\* placeholders with your real values before going live.

### 3.1 Global Settings

At the top of config.js, a few important global settings:

- units: "imperial" — Fahrenheit, mph, etc.
- timeFormat: 12 — 12-hour clock
- customCss: "css/custom.css" — points to our layout file (we'll create this in Section 4)
- language: "en", locale: "en-US"

### 3.2 Clock Module

```
{ module: "clock", position: "top_right", config: { displayType: "digital", displaySeconds: false } }
```

### 3.3 Weather Module

Using openmeteo — no API key required. Set your lat/lon to your location (this config is set to the Kansas City area):

```
{ module: "weather", position: "top_right", config: { weatherProvider: "openmeteo", type: "daily", maxNumberOfDays: 7, lat: 39.28, lon: -94.75 } }
```

### 3.4 Default Calendar Module (Data Source for CalendarExt3)

Keep the built-in calendar module in the config even though we're hiding it visually. CalendarExt3 reads its data from this module. Add your Google Calendar ICS URLs:

```
{ module: "calendar", header: "calendar", position: "bottom_center", classes: "cx3-theme overlay-calendar", config: { calendars: [ { name: "my_google", fetchInterval: 24*60*60*1000, symbol: "calendar-check", url: "https://calendar.google.com/calendar/ical/YOUR_URL.ics" } ] } }
```

 To get your ICS URL: Google Calendar > Settings > your calendar > Scroll to 'Secret address in iCal format'. Add one block per calendar you want to display.

### 3.5 MMM-CalendarExt3

This is what renders the actual grid in the bottom bar. It references the calendar data by the name you set above (my\_google):

```
{ module: "MMM-CalendarExt3", position: "bottom_bar",
  classes: "cx3-theme", title: "Monthly Calendar",
  config: { mode: "week", weekIndex: 0, weeksInView: 2,
    instanceId: "basicCalendar", maxEventLines: 7,
    firstDayOfWeek: 1, calendarSet: ['my_google'],
    fontSize: "24px", eventHeight: "26px",
    width: "800px" } }
```

### 3.6 MMM-BackgroundSlideshow

Sits in the fullscreen\_below position — behind everything else. The slideshowSpeed is in milliseconds (180000 = 3 minutes per image):

```
{ module: 'MMM-BackgroundSlideshow', position: 'fullscreen_below',
  config: { imagePaths: ['modules/MMM-BackgroundSlideshow/kitchenpiimages/'],
    randomizeimageOrder: true, slideshowSpeed: 180000 } }
```

### 3.7 MMM-MyScoreboard

Top left position. viewStyle: stackedwithlogos shows team logos. rolloverHours: 14 means yesterday's final scores stay visible until 2pm the next day, so you don't wake up to a blank board:

```
{ module: "MMM-MyScoreboard", position: "top_left", rolloverHours: 14,
  config: { viewStyle: "stackedwithlogos", showPlayofStatus: true,
    sports: [
      { league: "NHL", teams: ["BOS", "DET"] },
      { league: "NBA", teams: ["BOS"] },
      { league: "NFL", teams: ["NE", "KC"] },
      { league: "MLB", teams: ["BOS", "KC"] },
      { league: "NCAAM", teams: ["KU"] },
      { league: "NCAAF", teams: ["KU"] }
    ] } }
```

### 3.8 MMM-YouTubeSubscribers

Replace the \*\*\* values with your real API key and channel ID from Google Cloud Console:

```
{ module: 'MMM-YouTubeSubscribers', position: 'top_right',
  config: { apiKey: "YOUR_API_KEY",
    channelId: "YOUR_CHANNEL_ID",
    updateInterval: 3000000 } }
```

## 4. Custom CSS Layout (css/custom.css)

 Create this file at ~/MagicMirror/css/custom.css — it's referenced in config.js via the customCss setting we set earlier.

### 4.1 What We're Changing and Why

MagicMirror's default layout uses relative positioning and region padding that doesn't suit a mirror meant to be viewed from across a kitchen. Here's a summary of every change made:

## Base Font Size

Default is 16px. We bump it to 18px so text reads clearly at distance:

```
html { font-size: 18px; }
```

## Bottom Bar — Full Bleed Black Region

We absolutely position the `bottom_bar` to cover the bottom 50vh of the screen, extending it edge-to-edge by compensating for MagicMirror's built-in region padding with negative offsets:

```
.region.bottom_bar {  
  position: absolute; bottom: -10vh; left: -3.1vw;  
  width: 100vw; height: 50vh;  
  background: #000000; padding: 0; margin: 0;  
  box-sizing: border-box;  
}
```

 The negative bottom/left values are specific to this display. If you're on a different screen you may need to tweak these.

## Forcing CalendarExt3 Cells to Black

CalendarExt3 ships with its own CSS variables for cell backgrounds. Simply setting the region background to black isn't enough — the calendar cells render over it with their own gray/themed backgrounds. We have to override every layer:

- Set `--cellbgcolor: #000000` on the CX3 container
- Force `.cell`, `.thisMonth`, `.notThisMonth` to `background: #000000 !important`
- Force `.grid` and `.cellContainer` to `background: transparent !important`
- Force weekday header rows to `background: #000000 !important`

This was the most iterative part of the CSS — you'll notice some duplicate rules in the file from successive attempts to plug each bleed-through.

## Hiding the Default Calendar Module Visually

Because CalendarExt3 reads data from the default calendar module, we can't remove it from `config.js` — but we don't want it rendering its own event list on screen. Solution:

```
.module.calendar { display: none !important; }
```

## Weather Module — Black Pill, No Fading

Positioned tight to the top-right corner with a rounded black background (`border-radius: 25px`). MagicMirror's default `.dimmed` class fades certain weather elements — we override this:

```
.region.top.right .module.weather { background: #000000; border-radius: 25px; padding:  
8px 12px; }  
.region.top.right .module.weather,  
.region.top.right .module.weather .dimmed,  
.region.top.right .module.weather .bright {  
  opacity: 1 !important; color: white !important;  
}
```

## Background Slideshow Offset

The slideshow sits in `fullscreen_below` but is intentionally shifted right (left: 400px) and up (top: -100px) so it doesn't sit behind the scoreboard on the left side — keeping the text there legible against a darker background:

```
.region.fullscreen.below { position: absolute; top: -100px; left: 400px; width: 50vw; height: 70vh; }
```

## 4.2 CSS Typo Callouts

⚠️ **Two typos exist in the original CSS file that silently fail — worth showing on camera as a debugging lesson:**

### Typo 1: Invalid Hex Color — #0F522BA

Hex colors are 6 digits (#RRGGBB). #0F522BA has 7 digits — the browser silently ignores it and no color is applied. This was used for the calendar date numbers and event text color. The intended color was almost certainly:

```
color: #0F52BA; /* Sapphire blue — a valid 6-digit hex */
```

Or possibly #52BAxx — check against what color you were going for. Fixing this will actually apply a color to those calendar date numbers, which may or may not have been noticeable on your display.

### Typo 2: Invalid Font Size — 1.em

font-size: 1.em is not valid CSS — there's a missing number before the decimal. The browser ignores it. Valid options would be:

```
font-size: 1em; /* same as parent */  
font-size: 1.2em; /* 20% larger than parent */
```

Since this was applied to the `cellDate` (day-of-month number), 1em or 1.1em would keep it proportional to the `fontSize` set in the `CalendarExt3` config (24px).

## 5. TV Power Control via CEC

🔧 *This section has the most live troubleshooting story — walk through the problem you ran into with each approach before landing on the final solution.*

### 5.1 Install cec-utils

```
sudo apt update && sudo apt install cec-utils -y
```

Verify it can see your TV:

```
echo 'scan' | cec-client -s -d 1
```

You should see your TV listed as a device on the CEC bus. If nothing shows up, check that CEC is enabled in your TV's HDMI settings (sometimes called AnyNet+, Bravia Sync, SimpLink, etc. depending on brand).

### 5.2 The Problem with a Simple 'on 0'

The naive approach to turning on the TV is just:

```
echo 'on 0' | cec-client -s -d 1
```

This works — sometimes. But there are two failure modes you'll hit in the real world:

- The TV wakes from standby but doesn't switch to the Pi's HDMI input, landing on whatever it was last on
- cec-client hangs indefinitely if the TV doesn't respond, freezing your cron job

## 5.3 The Working `tv_on.sh` — Step by Step

Here's the final script with an explanation of every line:

```
#!/bin/bash
set -e
```

`set -e` exits immediately if any command fails — stops the script from continuing into a broken state.

```
timeout 8s bash -lc 'echo "on 0" | cec-client -s -t p -d 1'
```

Sends the power-on command to device 0 (the TV). The `timeout 8s` wrapper kills `cec-client` if it hangs for more than 8 seconds — critical for cron reliability. `-t p` sets the device type to playback device.

```
sleep 8
```

The TV needs time to fully boot its CEC stack before it will accept further commands. Without this sleep, the next commands are ignored. 8 seconds was the reliable threshold found through testing.

```
timeout 8s bash -lc 'echo "as" | cec-client -s -t p -d 1'
```

'as' = Active Source. Tells the TV that this HDMI input wants to be active. This should trigger an input switch — but on some TVs it still doesn't reliably work alone.

```
sleep 2
```

Small buffer before the final routing command.

```
timeout 8s bash -lc 'echo "tx 1F:82:10:00" | cec-client -s -t p -d 1'
```

This is the nuclear option — a raw CEC frame that explicitly sets the active HDMI routing path to port 1.0. The bytes break down as:

- 1F — broadcast from this device to all
- 82 — Active Source message
- 10:00 — physical address 1.0.0.0 (HDMI port 1)

This was the final fix that made input switching reliable. If your Pi is on a different HDMI port, you'll need to adjust the physical address.

## 5.4 `tv_off.sh`

```
#!/bin/bash
echo 'standby 0' | cec-client -s -d 1
```

Sends the standby command to the TV. Note the `-d l` flag here uses lowercase L (log to syslog) rather than `-d 1` (debug level 1) as in the on script — a minor inconsistency worth flagging on camera, but functionally fine.

## 5.5 Make Scripts Executable

```
mkdir -p /home/kitchenpi/scripts
nano /home/kitchenpi/scripts/tv_on.sh    # paste the script
nano /home/kitchenpi/scripts/tv_off.sh  # paste the script
chmod +x /home/kitchenpi/scripts/tv_on.sh
chmod +x /home/kitchenpi/scripts/tv_off.sh
```

Test them manually before setting up cron:

```
/home/kitchenpi/scripts/tv_on.sh
/home/kitchenpi/scripts/tv_off.sh
```

# 6. Scheduling with Cron

## 6.1 Open the Crontab Editor

```
crontab -e
```

If prompted, choose nano as the editor.

## 6.2 The Schedule

Add the PATH line first — this is critical. Without it, cron runs in a minimal environment and can't find cec-client:

```
PATH=/usr/bin:/bin
```

Then add the four schedule lines:

```
# Morning ON at 6:45 AM
45 6 * * * /home/kitchenpi/scripts/tv_on.sh

# Morning OFF at 8:00 AM
0 8 * * * /home/kitchenpi/scripts/tv_off.sh

# Evening ON at 4:30 PM
30 16 * * * /home/kitchenpi/scripts/tv_on.sh

# Evening OFF at 9:00 PM
0 21 * * * /home/kitchenpi/scripts/tv_off.sh
```

## 6.3 Reading the Cron Syntax

The five fields before the command are: minute hour day-of-month month day-of-week. An asterisk (\*) means 'every'. So:

- 45 6 \* \* \* = at minute 45 of hour 6, every day → 6:45 AM daily
- 0 8 \* \* \* = at minute 0 of hour 8, every day → 8:00 AM daily
- 30 16 \* \* \* = at minute 30 of hour 16, every day → 4:30 PM daily



- 0 21 \* \* \* = at minute 0 of hour 21, every day → 9:00 PM daily

## 6.4 Verify Cron is Running

```
| sudo systemctl status cron
```

Should show active (running). To check if your jobs ran:

```
| grep CRON /var/log/syslog | tail -20
```

## 6.5 Troubleshooting Cron + CEC

**⚠ Common issue: the script works when you run it manually but not from cron.**

This is almost always a PATH problem. Cron's default PATH doesn't include /usr/bin, which is where cec-client lives. That's why the PATH=/usr/bin:/bin line at the top of the crontab is essential.

Secondary issue: cec-client hanging in cron with no terminal attached, locking up the schedule. That's why every cec-client call in tv\_on.sh is wrapped in timeout 8s.

## 7. Final Checklist

 Run through this before calling the build done.

- MagicMirror starts on boot via PM2
- All modules visible: clock, weather, scoreboard, calendar grid, background slideshow, YouTube count
- Custom CSS file exists at ~/MagicMirror/css/custom.css and is referenced in config.js
- Calendar ICS URLs are live and CalendarExt3 is showing events
- tv\_on.sh and tv\_off.sh tested manually — TV turns on and switches to Pi input
- Crontab saved, PATH line present, cron service is active
- CSS typos optionally corrected: #0F52BA instead of #0F522BA, and a valid em value for font-size

— End of Guide —